

Applied NLP

DATA 641

Lecture 3 — NLP Pipeline; Feature Engineering

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Feature Engineering

- Feature engineering refers to the set of methods that will accomplish this task. It is also referred to as **feature extraction**.
- The goal of feature engineering is to capture the characteristics of the text into a numeric vector that can be understood by the ML algorithms.
- Some feature engineering approaches include approaches based on
 - classical NLP and traditional ML pipelines
 - DL pipelines

Classical NLP and traditional ML pipelines

Handcrafted features is a typical example for traditional feature engineering approach.

For example, imagine a task of sentiment classification on product reviews in e-commerce. One way to convert the reviews into meaningful "numbers" that helps predict the reviews' sentiments (positive or negative) would be to count the number of positive and negative words in each review.

One of the advantages of handcrafted features is that the model remains interpretable—it's possible to quantify exactly how much each feature is influencing the model prediction.

Approaches based on deep learning

The main drawback of classical ML models is the feature engineering. Handcrafted feature engineering becomes a bottleneck for both model performance and the model development cycle.

In the DL pipeline, the raw data (after preprocessing) is directly fed to a model. The model is capable of "learning" features from the data. Hence, these features are more in line with the task at hand, so they generally give improved performance.

But, since all these features are learned via model parameters, the model loses interpretability. It's very hard to explain a DL model's prediction.

Deep Learning

